

(A)symmetries in Kinyarwanda External Possession Constructions

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Abstract: This paper addresses the syntactic properties of two types of external possession constructions (EPCs) in Kinyarwanda. I propose both types involve an external possessor in the Spec of an ApplP. I show that alienable possessors move to the Spec of a HighApplP, while inalienable possessors are base-generated at SpecLowApplP. Asymmetries between the different types of EPCs are attributable to their unique derivations. This paper thus shows that even within a single language, different syntactic configurations can seemingly derive the same semantic interpretation.

1 External possession constructions in Kinyarwanda

Kimenyi (1980) describes two distinct patterns of external possession constructions (EPCs) in Kinyarwanda (JD61), a Bantu language of Rwanda. This paper explores two realizations of Kinyarwanda EPCs and accounts for their different distribution and properties. The first type of EPC, shown in (1), requires no overt valency changing morphology shown on the verb in (1b) for the external possessor to appear syntactically independent of the possessum. In the second type, shown in (2), the realization of external possessor as a verbal dependent in (2a) requires the verb obligatorily bear an applicative suffix not seen in (2b).

- (1) a. umugóre y-a-shokoj-e umusatsi w'=ûmugabo
 woman SM₁-PST-comb-ASP hair CON₃=man
- b. umugóre y-a-shokoj-e umugabo umusatsi
 woman SM₁-PST-comb-ASP man hair
 'The woman combed the man's hair.'
 lit. 'The woman combed the man the hair.'
- (2) a. Ingurube z-a-ri-iye ibíryo by'=abaána
 pigs 10SM-PST-eat-ASP food CON₈=children
- b. Ingurube z-a-ri-ir-iye abaána ibíryo.
 pigs 10SM-PST-eat-APPL-ASP children food

‘The pigs ate the children’s food’
 lit. ‘The pigs ate children the food’
 (Kimenyi 1980:p. 105)

Adopting the terminology of Van de Velde (2020), I refer to the first type in (1) as *nuclear EPCs*, and to the second type in (2) as *applied EPCs*. Nuclear EPCs require a part-whole relationship between the possessor and possessum, while applied EPCs involve no particular relation beyond possession (Kimenyi 1980). Both EPCs show different syntactic properties and vary in object symmetry. Only in nuclear EPCs do both possessor and possessum retain prototypical objecthood properties. In (3), both can become the subject of a passive clause, and both can be pronominally incorporated on the verb, as shown in (4).

- (3) a. Umusatsi w-a-shokoj-w-e umugabo n’úmugore
 hair 3SM-PST-comb-PASS-ASP man by the woman
- b. Umugabo y-a-shokoj-w-e umusatsi n’úmugore
 man 1SM-PST-comb-PASS-ASP hair by the woman
 ‘The man’s hair was combed by the woman’
- (4) a. Úmugore y-a-wu-shokoj-e umugabo
 woman 1SM-PST-3OM-comb-ASP man
 ‘The woman combed it (the hair of) the man’
- b. Úmugore y-a-mu-shokoj-e umusatsi
 woman 1SM-PST-1OM-comb-ASP hair
 The woman combed him the hair (of the man)’
 (Kimenyi 1980:p. 104)

In contrast, applied EPCs involve an apparent demotion of the possessum so it no longer behaves as DO. The possessum cannot be passivized in (5) or incorporated in (6).

- (5) a. * Ibíryo by-a-ri-ir-iw-e abaána n’íngurube
 food 8SM-PST-eat-APPL-PASS-ASP children by pigs
- b. Abaána y-a-ri-ir-iw-e ibíryo n’íngurube
 children 1SM-PST-eat-APPL-PASS-ASP food by pigs
 ‘The children’s food was eaten by the pigs.’
- (6) a. * Íngurube z-a-bi-ri-ir-e abaána
 pigs 10SM-PST-8OM-eat-APPL-ASP children
- b. Íngurube z-a-ba-ri-ir-e ibíryo
 pigs 10SM-PST-2OM-eat-APPL-ASP food
 ‘The pigs ate it (the children’s) food’
 (Kimenyi 1980:p. 104)

Deal (2017) notes there is no uniform cross-linguistic syntactic encoding of external possession, and Tyler (2021) argues that even a single language may have multiple means of deriving external possession, as in the case of Choctaw. PCs in Kinyarwanda further this proposal and

demonstrate that EPCs may be epiphenomenal. Beyond the part-whole requirement on nuclear EPCs, applied and nuclear EPCs appear semantically similar; both express that the possessor is affected in some way by the action taken upon the theme (Van de Velde 2020, Kimenyi 1980). My analysis provides an account of the observed differences in object (a)symmetries and other syntactic behaviours between nuclear and applied EPCs, demonstrating they are derived via different means.

The structure of this paper is as follows. The next section overviews previous and potential alternative accounts of external possession in Kinyarwanda and underscores their shortcomings. Section 3 outlines my proposal and attributes syntactic properties of EPCs in Kinyarwanda to different positions of applied and nuclear external possessors. Section 3.1 argues for possessor raising in applied EPCs over other theories. In contrast, I propose that nuclear EPCs may be base-generated. Section 4 fleshes out the analysis and provides a phase-based account of unresolved syntactic properties of each EPC, strengthening its explanatory power.

2 Background

I overview here some background and potential alternative analyses of EPCs in Kinyarwanda that veer from the raising and LowAppl account I propose in section 3. There are three main alternatives I cover. First, I consider analyses put forth by Bickford (1986) and Davies (1997) of Kinyarwanda EPCs within a Relational Grammar framework. Although much of the data can be accounted for, neither RG account sufficiently explains the restrictions on =A-movement in applied EPCs. In section 2.2, I consider a proposal from Simango (2007) that applied EPCs in Chichewa do not actually express possession and show it cannot be applied to Kinyarwanda. Lastly, in section 2.3, I consider that Kinyarwanda EPCs could be derived via binding. Although this may initially have merits, it does not account for the lack of reconstruction effects and appearance of subject reflexive possessors.

2.1 Possessor ascension

Using data from Kimenyi's (1980) general analysis of grammatical relations in Kinyarwanda, the first focused account of external possession in Kinyarwanda is given by Bickford (1986) within a Relational Grammar framework. Bickford follows Dryer (1983) and opposes Gary and Keenan's (1977) proposal that Kinyarwanda does not differentiate DO and IO, arguing that Kinyarwanda distinguishes between three different initial object term relations; DO, IO, and Ben. He argues nuclear EPCs involve promotion of the possessor to IO while maintaining the possessum as DO. Applied EPCs, in his analysis, promote the possessor to DO and put the possessum *en chômeage*, stripping it of any grammatical relation¹. His primary aim is to show that Kinyarwanda does, in fact, distinguish between IO and DO, contra Gary and Keenan (1977), on the basis of object asymmetries in EPCs.

¹Bickford and Davies (1997) use the terms alienable possessor ascension and inalienable possessor ascension to refer to what I call applied and nuclear EPCs respectively.

The most recent address of EPCs in Kinyarwanda, however, comes from Davies (1997). Davies primarily argues against Bickford's analysis, and gives an alternative relation arc. Davies instead proposes that, instead of being promoted directly to DO, applied external possessors are promoted to the grammatical relation of whichever host they ascend from (i.e. the possessor always bear the grammatical relation the possessum would bear if there were internal possession). Davies also assumes that any instance of applied possessor ascension puts the host *en chômage*.

Accounting for applied EPCs, Davies proposes the constraint in (7), which effectively guarantees that any ascended possessor in an applied EPC is always either subject or direct object. This is meant to account for the applied external possessor always having object properties, as noted above.

(7) *Final Nuclear Term Constraint* (Davies 1997:p. 97)

An ascension arc in the APA structure must be a nuclear term arc (i.e., with R-sign 1 or 2) in the final stratum.

This constraint accounts for the fact that there cannot be multiple applied external possessors, shown in (8), as doing so would necessarily demote one from 1 or 2 (i.e. SU or DO), and any final stratum where an applied external possessor is neither 1 or 2 is a violation of the Final Nuclear Term Constraint in (7).

- (8) * Umukoôbwa a-ra-som-er-er-a umugóre umwáana igitabo.
 girl 1SM-PRES-read-APPL-APPL-ASP woman child book
 Intended: 'The girl is reading the woman's child's book.'
 (Davies 1997:p. 99)

I do not delve too deeply into the arguments presented by Davies in favour of his analysis. The RG account of possessor ascension as proposed by Davies is able to account for most if not all of the properties of object asymmetry in applied EPCs involving A-movement (i.e. incorporation, passivization). However, one apparent issue with the RG account of applied EPCs is that we are left unable to explain why the possessum in applied EPCs cannot undergo $\bar{A}$ -movement, such as relativization (9) or clefting, while the nuclear EPC possessum may.

- (9) a. ábáana ingurube z-aa-ri-ír-iye ibíryo
 children pigs 10SM-PST-REL.eat-APPL-PFV food
 'The children that the pigs ate the food of.'
 b. * ibíryo ingurube z-aa-ri-ír-iye ábáana
 food pigs 10SM-PST-REL.eat-APPL-PFV children
 Intended: 'The food that the pigs ate of the children.'
 (Davies 1997:p. 92)

Meanwhile, English agentive obliques in passives can undergo relativization, as in (10), despite being assumed to be put *en chômage*.

- (10) The man that the food was eaten by.

Without proposing an additional Kinyarwanda-specific constraint within RG preventing chômeurs from undergoing relativization, the RG analysis misses an account of $\bar{A}$ -movement. In section 4, I propose a phase-based analysis to account for the inability of the applied possessum to undergo $\bar{A}$ -movement.

2.2 Non-possession

Simango (2007) presents an analysis of apparent EPCs in Chichewa (N30). A central idea of theirs is that Chichewa nuclear and applied EPCs are unrelated constructions. He argues applied EPCs are not actually EPCs in Chichewa, but are properly analyzed as benefactive constructions, where applicative morphology solely introduces a beneficiary argument². The possessive reading of such sentences are reduced to an implicature. When an entity is affected by another entity being acted on, the affected entity is most often in possession of the entity which is acted upon. I do not contend with the arguments made for this analysis in Chichewa, but I argue this cannot be the case in Kinyarwanda applied EPCs. Below, I overview one of Simango’s main arguments for this in Chichewa, and show it does not hold for Kinyarwanda.

Simango puts forward sentences with co-occurrence of applied arguments and internal possession as evidence for his claim Simango (2007:p. 936). Observe the following sentences from Chichewa.

- (11) a. John a-na-ndi-masul-ir-a zipi wa-nga
 John SM-PST-1SG.OM-unfasten-APPL-FV zipper ASSOC-1SG.POSSESS
 ‘John unfastened my zipper for me’
- b. John a-na-ndi-masul-ir-a zipi wa-ke
 John SM-PST-1SG.OM-unfasten-APPL-FV zipper ASSOC-3SG.POSSESS
 ‘John unfastened his zipper for me’
 (Chichewa, N30) (Simango 2007:p. 935)

According to Simango’s argumentation, the possessor in both (11a) and (11b) can be realized as an internal possessor because the applied argument does not specify the possessor. Regarding the sentences, he states that “...the introduction of another expression to specify the possessor would not be possible if the said possessor had already raised and surfaced as an object prefix on the verb, triggering the applicative suffix in the process.” (Simango 2007:p. 935). This predicts that, if the applied argument were truly a possessor of the, it could not co-occur with internal possession. In Kinyarwanda, this is exactly the case.

The comparable Kinyarwanda sentences below are unexplained if the applied argument does not denote a possessor, as may be the case in Chichewa. Notice the change in acceptability associated with the co-occurrence of applicative morphology, reflexivization, and internal possession.

²I will continue to refer to the comparable Chichewa constructions as applied EPCs for reasons of clarity, although I am agnostic to their best analysis in Chichewa,

- (12) a. Abagabo ba-r-uúbak-a inzu yaá bo
 men 2SM-PRES-build-ASP house of them
 ‘The men are building their house.’
- b. Abagabo ba-r-íy-uúbak-ir-a inzu
 men 2SM-PRES-REFL-build-APPL-ASP house
 ‘The men are building their own house.’
- c. * Abagabo ba-r-íy-uúbak-ir-a inzu yaá bo
 men 2SM-PRES-REFL-build-APPL-ASP house of them
 Intended: ‘The men are building their own house.’
 (Kimenyi 1980:p. 57)

In (12a), the theme argument *inzu* is internally possessed by *yaá bo*. An applied EPC is shown in (12b). The applicative suffix *-ir/-er* appears on the verb, and the applied argument is realized as a reflexive clitic pronoun on the verb with *íy-*. However, (12c) is made ill-formed with co-occurrence of the applied reflexive clitic and internal possession, where the genitive internal possessor and applied reflexive are co-referential. This ill-formedness is explained as a result of apparent double-marking of possession. If the applicative suffix *-ir/-er* solely expresses affect- edness and not possession, the ungrammaticality of (12c) is unexpected. These data show that applied EPCs are, at the very least, more closely related to internal possession in Kinyarwanda than in Chichewa.

2.3 Binding

A potential account of external possession in Kinyarwanda is that the possessive interpretation is derived via binding. In this analysis, as outlined by Deal (2017), we could imagine two co-referential arguments; one possessor and one marked as affectee or some other thematic role. There is some level of explanatory power here. Observe the sentences in (13).

- (13) a. Úmwaána y-iicar-i-yé-ho íntebe umugabo
 child 1SM-sit-APPL-ASP-LOC chair man
 ‘The child is sitting in the chair for the man’
 ‘*The child is sitting in the man’s chair’
- b. Úmwaána y-iicar-i-yé-ho umugabo íntebe
 child 1SM-sit-APPL-ASP-LOC man chair
 ‘The child is sitting in the man’s chair’
 ‘The child is sitting in the chair for the man’
 (McGinnis and Gerds 2004:p. 163)

I address this more thoroughly in section 3.1.1, but the noteworthy property is that the possessor must precede the possessum for the external possession reading. Otherwise, *umugabo* is interpreted solely as a beneficiary in (13a).

It may be argued that the unavailability of the possessive reading in (19a) results from Condition A of the Binding Theory. Say there is some null internal possessor PRO that becomes

co-referential with the applied argument via c-command. In such an analysis, *umugabo* cannot be interpreted as the possessor of *intebe* in (13a) as the sentence's linear order suggests the former noun does not c-command the latter. I discuss issues with this analysis here.

2.3.1 Lack of reconstruction

While binding may adequately explain the lack of possessive interpretation in (13a), like the RG account, it struggles to explain $\bar{A}$ -movement of the possessum, as in cases of left dislocation, seen below.

- (14) a. Yohaâni y-a-som-e-ye Maríyai igitaboi
 John 1SM-PST-read-APPL-ASP Mary book
 'John is reading Mary's book.'
- b. Maríyai, Yohaâni y-a-mu-som-e-ye igitaboi
 Mary John 1SM-PST-1OM-read-APPL-ASP book
 'Maryi, John is reading heri book.'
- c. * Igitaboi, Yohaâni y-a-gii-som-e-ye Maríyai
 book John 1SM-PST-7OM-read-APPL-ASP Mary
 Intended: 'Mary's book, John is reading it.'

The lack of possessive reading of (14c) is unexpected under a binding analysis. A binding analysis correctly predicts that when both *Mariya* and *igitabo* are in A-positions, the former must c-command the latter for the possession reading. However, it does not account for the inaccessibility of the possessive reading in (14c), when *igitabo* has $\bar{A}$ -moved. Being in an $\bar{A}$ -position, *igitabo* is expected to be able to undergo LF-reconstruction, as in the English sentences in (15).

- (15) a. Which picture of himselfi did Johni like?
 b. That picture of himselfi, Johni really liked it.

If binding did derive external possession in (14a), we would expect the possessor reading to persist in (14c) after LF-reconstruction, as seen in English sentences. The lack of such interpretation despite the availability of reconstruction suggests binding does not derive external possession.

2.3.2 Subject possessors

Additional evidence against a binding analysis of external possession is subject possessors. Consider the sentences with reflexive object markers in (16) and (17).

- (16) a. Umugabo y-a-som-ye igitabo.
 man 1SM-PST-read-ASP book
 'The man is reading the book'

- b. Umugabo y-a-í-som-e-ye igitabo.
 man 1SM-PST-REFL-read-APPL-ASP book
 ‘The man is reading his own book’
 (Kimenyi 1980:p. 149)
- (17) a. Abaána ba-ra-shyir-a ibi ruhaánde rw-aá=bo
 children 2SM-PRES-put-ASP books near of them
 ‘The children are putting books near them.’
- b. Abaána ba-r-í-shyir-a ibi iruhaánde
 children 2SM-PRES-REFL-put-ASP books at side
 ‘The children are putting books at their side.’
 (Kimenyi 1980:p. 103)

In (16b), the applied external possessor surfaces as a reflexive object marker on the verb. In (17b), we see the same with a nuclear external possessor, the difference being the lack of overt applicative morphology. In both sentences, the reflexive object marker is the possessor of the theme and is co-referential with the subject of the sentence. This is not ideal for a binding analysis of Kinyarwanda EPCs.

If binding were all that is needed for external possession in Kinyarwanda, we expect that the possessive reading should occur without need for any reflexive pronoun on the verb, as the subject already c-commands the theme. The need for the reflexive pronoun to indicate an external possessor co-referential with the subject provides more evidence against a binding account.

2.3.3 Passives

One last issue with a binding account of external possession in Kinyarwanda is passive clauses. Recall that in (4) with nuclear EPCs, both nuclear possessor and possessum can raise to become the subject of a passive clause. In (18), we see the possessum serve as a passive subject, while the possessor remains lower in the clause.

- (18) Umusatsi w-a-shokoj-w-e umugabo n’úmugore
 hair 3SM-PST-comb-PASS-ASP man by the woman
 ‘The man’s hair was combed by the woman’

A binding account of external possession cannot account for the fact that *umugabo* is still understood as the possessor of *umusatsi* in (18). Under a binding analysis, a possessive reading would be unexpected in (18) since the possessum now asymmetrically c-commands the possessor.

3 Derivation of EPCs

In this section I address the derivation of both types of EPCs, beginning with applied EPCs. I propose applied external possessors are base-generated DP-internally with the possessum and

applied EPCs derived via raising to the Spec of a HighApplP. I motivate the raising analysis on the basis of word order constraints and adjunct island effects. In section 4, I show the raising analysis accounts for object asymmetries in applied EPCs. Regarding nuclear EPCs, I propose the possessor is base-generated in the Spec of a VP-internal LowApplP, requiring no movement.

3.1 Possessor raising

I argue here that applied EPCs are best analyzed as derived from underlying internal possession. Under my analysis, the possessor is base-generated within the possessum DP and raises out to a higher functional projection, which I assume to be SpecHighApplP. Two primary facts about Kinyarwanda EPCs are furthered in support of raising; word order and island effects. That the landing site is a HighApplP is motivated by apparent homophony and external possession of applied instrumental arguments.

3.1.1 Word order

The first piece of evidence in favour of a raising analysis for Kinyarwanda EPCs comes from a restriction put on the possible ordering of arguments in EPCs. Recall the sentences in (13b), repeated as (19) below.

- (19) a. Úmwaána y-iicar-i-yé-ho ítebe umugabo
 child 1SM-sit-APPL-ASP-LOC chair man
 ‘The child is sitting in the chair for the man’
 ‘*The child is sitting in the man’s chair’
- b. Úmwaána y-iicar-i-yé-ho umugabo ítebe
 child 1SM-sit-APPL-ASP-LOC man chair
 ‘The child is sitting in the man’s chair’
 ‘The child is sitting in the chair for the man’

In (19b), the sentence’s reading is ambiguous regarding the man’s status as the possessor of the chair or beneficiary of the event, without any direct relation to the chair. In (19a), only the latter reading is available. The man cannot be understood as the possessor of the chair in this sentence.

The apparent takeaway is that, for the applied argument to be the possessor of another DP in the clause, the possessor must precede the possessum. With a reversed order, the possession reading is unavailable. There are two potential explanations that might properly account for this fact. In the first, we may claim that *umugabo* in (19b) is base-generated outside of the possessum DP. This derives the binding analysis discussed above, where the possessor’s c-command of a null PRO in the specifier of the possessum DP, where PRO receives a possessor θ -role, allows for co-referentiality. The second possibility, which I argue for, is that *umugabo* in (19b) is base-generated within the possessum DP and raises to an external position. The word order requirement then follows from the fact that the possessor must be in a position above the possessum as it has raised from there.

3.1.2 Island effects

More apparent evidence of possessor raising comes from apparent island extraction effects. The set of sentences below show that applied external possession seems to only be allowed where extraction is allowed.

- (20) a. * Umuhuûngu y-a-andik-i-ye íbarúwa umukoôbwa n'=íkarámu.
 boy 1SM-PST-write-APPL-ASP letter girl with=pen
- b. Umuhuûngu y-a-andik-iish-i-ye umukoôbwa íbarúwa íkarámu.
 boy 1SM-PST-write-INSTR-APPL-ASP girl letter pen
 ‘The boy wrote the letter with the girl’s pen.’
 (Kimenyi 1980:p. 112)

In (20a), the intended possessor *umukoôbwa* cannot appear independent of the intended possessum *íkarámu*. The sentence is ill-formed with an intended possession reading. In (20b), with the same possessor and possessum, external possession is licensed and the resulting sentence is well-formed with the intended meaning. The crucial difference in these sentences is that in (20b) the instrumental possessum is syntactically encoded as an argument. In (20a), the instrumental possessum DP is syntactically encoded as an adjunct, introduced by a PP. It is with the instrumental suffix in (20b) that the instrumental possessum DP is able to be realized as an argument rather than an adjunct.

We can understand this restriction as a result of an adjunct extraction island. It is not possible for the possessor to raise unless the instrument is properly realized as a verbal argument through the use of the instrumental suffix, as in (20b). As long as the instrumental DP is realized with instrumental applicative morphology, the possessor is able to act as DO. If the instrumental is realized as an adjunct, the possessor cannot raise to escape that adjunct PP.

3.1.3 Raising to SpecApplP

I propose that applied EPCs are derived via movement of the possessor DP to a position outside VP, the specifier of a HighApplP. This intuitively accounts for several easily identifiable properties of applied EPCs. Firstly, the word order facts are derived nicely. Given that the external possessor precedes all other arguments, it is expected that its position is above those other arguments in the syntax.

A second intuitive reason is that the morpheme used in applied EPCs is homonymous with the benefactive applicative suffix, and the two may not co-occur, as seen in (8). McGinnis and Gerdt (2004) claim this to be a high applicative in the sense of Pytkänen (2000), given its semantic interpretation of relating individuals to events. Furthermore, the benefactive applicative is able to combine with unergative verbs with no VP-complement, a standard test to determine between HighAppls and LowAppls³.

³I gloss this simply as APPL rather than BEN as they are the same within my analysis.

- (21) Umugóre a-rá-kor-er-a umugabo
 woman 1SM-PRES-work-APPL-ASP man
 ‘The woman works for the man.’
 (Kimenyi 1980:p. 15)

The fact that applied EPCs cannot appear with unergative verbs without an object is, however, not an issue with this analysis. This is a result of the simple inherent property of applied EPCs within my analysis as involving raising to SpecHighApplP. Assuming that the possessor is base-generated within the DP of a lower argument before raising to SpecHighApplP, there must be a lower argument to originate from in the first place.

There is additional evidence to suggest that the position of the external possessor should be in a HighApplP rather than LowApplP. Recall the sentence in (20b), which shows an applied EPC with an external possessor of an instrumental applied argument. Instrumental applicatives are also high, as can be observed by the fact that they occur with intransitive verbs as in (22).

- (22) Umwaána a-r’ a-rí-iish-a ikánya.
 child 1SM-PRES-eat-INST-ASP fork
 ‘The child is eating with a fork.’
 (Kimenyi 1980:p. 79)

For the applied external possessor to be the possessor of an applied instrumental argument, it must be structurally higher than the instrumental applicative. Therefore, the ApplP hosting the external possessor is necessarily above the instrumental HighApplP, and must also be high. This is additionally supported by the fact that, as shown in (20b), the APPL suffix follows the INST suffix, which appears closest to the root. If it were not the case that the possessor raised from the instrumental HighApplP to a higher HighApplP, we could not account for the adjunct island effects.

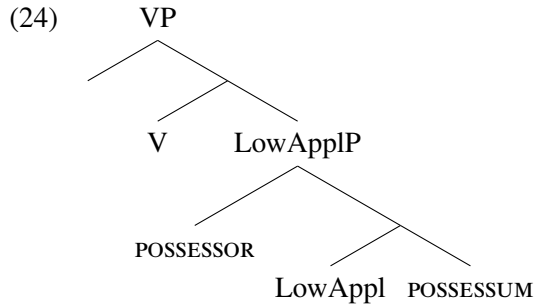
It is arguable whether the raising advocated here is thematic raising in the sense of Hornstein (1999) or athematic. On the one hand, both Kimenyi (1980) and Van de Velde (2020) note that the external possessor is given some thematic role of either affectee or concerne as well as possessor. On the other hand, there are some sentences involving stative verbs, such as (23), where it is not quite clear the degree to which the applied external possessor is affected.

- (23) Umukoôbwa a-be-er-eye umuhuûngu muu nzu.
 girl 1SM-BE-APPL-PFV boy in house
 ‘The girl is in the boy’s house.’
 (Kimenyi 1980:p. 100)

I do not have data to test if affectedness is a strict semantic requirement of applied external possessors. Despite thematic raising being more controversial, I simply assume that the possessor raises to a position where it receives an additional AFFECTEE θ -role due to lack of firm evidence for or against this analysis, though neither position can be falsified with existing data.

3.2 Low applicatives

As for nuclear EPCs, I argue that they are derived via a LowApplAT as proposed by Cuervo (2003). In this analysis, the possessor occupies the specifier of a LowApplP that expresses a static relationship to the complement DP, deriving the possessive meaning and giving the structure in (24) with the LowApplP as a complement of V. It is not necessary to propose a raising analysis for nuclear EPCs here, as whether the possessor is base-generated at SpecLowApplP or DP-internally and raises to that position makes no material difference to our analysis and the predictions are the same as long as the external possessor lands at SpecLowApplP⁴. In this analysis, nuclear external possessors are expected to pattern with indirect objects with canonical ditransitives, which is reminiscent of Bickford's (1986) claim that nuclear external possessors ascend directly to IO. I show below that this is borne out and offer additional evidence for their position below VP.



3.2.1 Object symmetries

As previously discussed, we can note that external possessors and possessa in nuclear EPCs behave symmetrically with respect to their ability to undergo passivization (25) and pronominal incorporation (26).

- (25) a. Umusatsi w-a-shokoj-w-e umugabo n'úmugore
hair 3SM-PST-comb-PASS-ASP man by the woman
- b. Umugabo y-a-shokoj-w-e umusatsi n'úmugore
man 1SM-PST-comb-PASS-ASP hair by the woman
'The man's hair was combed by the woman'
- (26) a. Úmugore y-a-wu-shokoj-e umugabo
woman 1SM-PST-3OM-comb-ASP man
'The woman combed it (the hair of) the man'
- b. Úmugore y-a-mu-shokoj-e umusatsi
woman 1SM-PST-1OM-comb-ASP hair
The woman combed the hair it (of the man)'
(Kimenyi 1980:p. 104)

⁴We could also claim that applied external possessors are base-generated at SpecLowApplP and raise to SpecHighApplP. This may make some slightly different predictions regarding their behaviour with double object constructions, but I do not have the data to argue for or against this, and as such assume they simply begin DP-internally.

Kimenyi (1980:p. 104) claims that, in fact, possessa in nuclear EPCs retain all of their DO properties except for the ability to undergo reflexivization or relativization. He claims that reflexivization cannot occur simply because the possessor and possessum are never identical. No examples disproving the possibility of relativization are given, but he claims it is a semantic/pragmatic effect of the possessum being unique that prevents it from undergoing relativization, rather than a syntactic restriction. Both possessor and possessum can be relativized in the case of exclusive insertion, as seen in (27).

- (27) a. Nta musatsi umugóre y-a-shókoj-e umugabo.
no hair woman 1SM-PST-REL.comb-PFV man
'The woman didn't comb the hair of the man at all.'
(lit. There is no hair that the woman combed the man) (Kimenyi 1980:p. 106)
- b. Nta mugabo umugóre y-a-shókoj-e umusatsi.
no man woman 1SM-PST-REL.comb-PFV hair
'There is no man that the woman combed the hair of.'
(Simon Pierre Munyaneza, personal communication)

Possessor and possessa in nuclear EPCs have the same object symmetry as recipient/goal and theme in double object constructions. Both are able to undergo relativization (28), incorporation (30), and passivization (29).

- (28) a. N-a-boon-ye igitabo umuhuûngu y-a-haá-ye umukoôbwa.
1SM-PST-see-PFV book boy 1SM-PST-REL.givePFV girl
'I saw the book that the boy gave to the girl.'
- b. N-a-boon-ye umukoôbwa umuhuûngu y-a-haá-ye igitabo.
1SM-PST-see-PFV book boy 1SM-PST-REL.givePFV girl
'I saw the girl that the boy gave the book to.'
(Kimenyi 1980:p. 62)
- (29) a. Ibitabo by-eerek-e-w-e umugóre abáana n'=úmugabo.
books 8SM-show-APPL-PASS-PFV woman children by=man
'The books were shown to the children for the woman by the man.'
- b. Abáana y-eerek-e-w-e umugóre ibitabo n'=úmugabo.
children 1SM-show-APPL-PASS-PFV woman books by=man
'The children were shown books for the woman by the man.'
(Kimenyi 1980:p. 136)
- (30) Umugóre y-a-by-íy-erets-e.
man 1SM-PST-8OM-REFL-show-PFV
'The woman showed them to herself.'
(Kimenyi 1980:p. 134)

These similarities with respect to object symmetry are expected if the possessor and possessum in nuclear EPCs occupy what is essentially structurally the same position as recipient/goal and theme respectively in DOCs. In the next section, I offer more evidence to further this position.

3.2.2 Co-occurrence

I have proposed the structure in (24) to account for nuclear EPCs. One prediction this analysis makes is that nuclear EPCs should not co-occur with canonical DOCs that do not involve external possession. Indeed, Davies (1997:p. 102) observes this is exactly the case, as shown in (31).

- (31) a. Umugóre á-r-éerek-a umuhuûngu amagaru y'=ûmûkoobwa.
 woman 1SM-PST-show-ASP boy legs CON₆=girl
 'The woman is showing the boy the girl's legs.'
- b. * Umugóre á-r-éerek-a umuhuûngu umûkoobwa amagaru.
 woman 1SM-PST-show-ASP boy girl legs
 Intended: 'The woman is showing the boy the girl's legs.'
- (Davies 1997:p. 102)

(31b) is ill-formed with a nuclear EPC in a DOC. Our analysis, as outlined above, explains this naturally. The possessor and possessum are introduced as the specifier and complement of a LowApplP. Given that this LowApplP must be a complement of V, there cannot be two. In a DOC, a LowApplP_{TO} complement of V occupies the same position that our LowApplP_{AT} would otherwise occupy. Their inability to co-occur supports our analysis.

3.2.3 Instrumental possessa

More evidence in favour of the nuclear external possessor being below VP is in the form of instrumental possessa. With applied EPCs, the fact that instruments could only be externally possessed when realized as applied arguments supports that applied external possessors are higher than the instrumental HighApplP. Following this logic, we expect that if nuclear external possessors are lower than VP, and thus below the instrumental HighApplP, we should never have nuclear external possession of an applied instrumental argument. The sentences below confirm this prediction in Kinyarwanda.

- (32) a. N-a-tet-se keke n'=amaboko yanjye
 1SM-PST-cook-PFV cake with=hands my
 'I baked the cake with my own hands.'
- b. N-a-tek-esh-eje keke amaboko yanjye
 1SM-PST-cook-INST-PFV cake hands my
 'I baked the cake with my own hands.'
- c. * N-a-n-tek-esh-eje keke amaboko
 1SM-PST-1SG.OM-cook-INST-PFV cake hands
 Intended: 'I baked the cake with my own hands.'
- (Simon Pierre Munyaneze, personal communication)

In (32b), internal possession of a body part is expressed with the first person possessive marker *yanjye*. In this example, the body part is an applied instrumental argument. Given that there is a part-whole relationship between the possessor and possessum, there is nothing semantic

that should prevent a nuclear EPC from being used here to express the possession relationship. That the instrumental applicative realizes the instrument as a full argument means there should be no island effect preventing external possession. However, in (32c), where the intended nuclear external possessor is incorporated as a pronoun on the verb, the result is ill-formed. This is explained if the nuclear external possessor is in the specifier of LowApplPAT, a VP complement. As such, (32c) is predicted to be ill-formed since the applied instrumental argument is base-generated at the specifier of a HighApplP above the VP and above LowApplPAT.

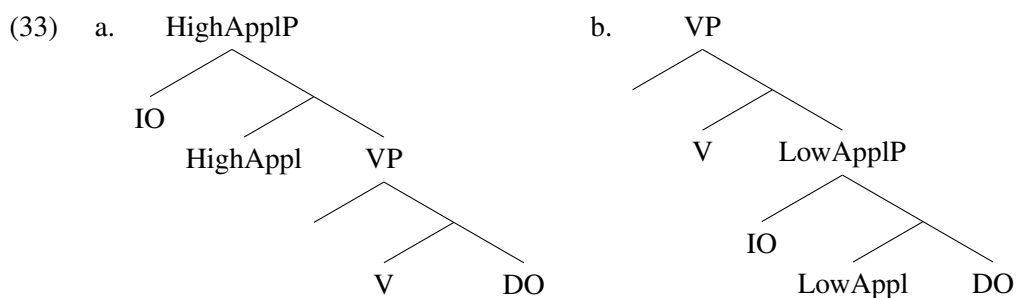
To summarize, nuclear EPCs are derived via LowApplAT on the basis of object symmetries shared with those of recipient/theme, their inability to co-occur with DOCs, and their inability to associate with arguments higher than VP. I have proposed that applied external possessors must be higher than VP due to their ability to possess arguments above VP, word order, and homonymy with the benefactive applicative suffix. The raising analysis proposed follows from the observed word order constraints and island effects. In the following section, I show that the differences in object symmetry between both types of EPCs fall from their respective derivations.

4 Asymmetries

Here, I propose that the syntactic differences between nuclear and applied EPCs can be attributed to the proposal by McGinnis (2000, 2001) that high applicatives are phases, while low applicatives are not. I briefly outline McGinnis' proposal and show that upon its adoption, the observed asymmetries between nuclear and applied EPCs can be explained by the respective derivations proposed above.

4.1 Applicative phases

McGinnis (2000, 2001) proposes that a large degree of linguistic variation in argument (a)symmetries is accounted for by differences in applicative phase structures. She notes two primary categories of applicatives, E-applicatives relates an event to an individual, and I-applicatives relate two individuals. These essentially correspond to Pylkkänen's (2000) HighApplP and LowApplP, respectively, and I will refer to them with these terms. HighApplP appears above VP, as in (33a), and LowApplP appears below VP as a complement of V, as in (33b).



McGinnis assumes that upon completion of a phase, the only accessible elements within the phrase that forms the phase are heads, adjuncts, or those in the specifier position, in accordance

with the Phase Impenetrability Condition (Chomsky and Kenstowicz 1999). Elements within the domain of the phase are inaccessible to any higher operations. An optional phase-EPP feature is allowed that triggers movement of an element within the phasal domain to its specifier, i.e. the escape hatch. Distinguishing between HighApplPs and LowApplPs, she makes the claim in (34).

- (34) The sister of VP heads a phase if it assigns a theta-role to a syntactic argument.
(McGinnis 2000:p. 5)

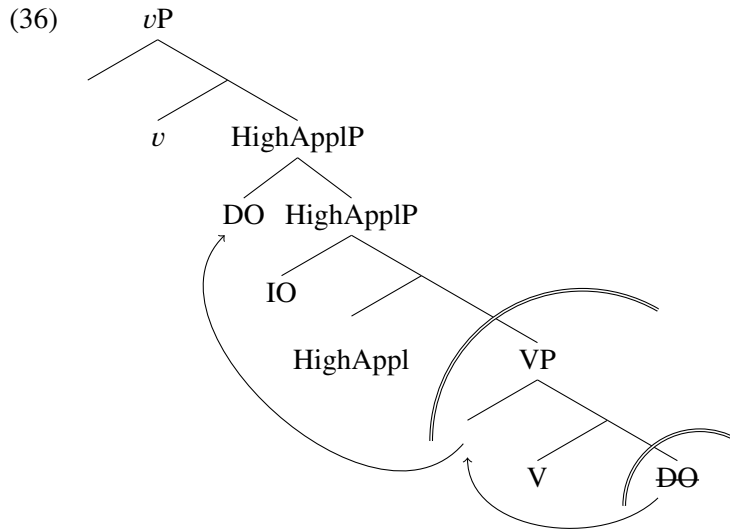
Given this claim, only HighApplP forms a phase, as it is a sister of VP while LowApplP is not. This accounts for object asymmetries in English DOCs. In such constructions, only the higher element, the IO, is able to raise to SpecVP to escape its phasal domain, while movement of the lower element, DO, to SpecVP would be a violation of Shortest Movement. Therefore, the argument in SpecLowApplP is predicted to have object properties, while the complement of LowApplP is trapped within the VP phase.

McGinnis and Gerdtz (2004) argue that object symmetries with high applicatives in Kinyarwanda can be derived as a result of HighApplP being phasal. In (35), both the applied benefactive argument (*umugabo*) and the applied locative argument (*íntebe*) are shown able to be the subject of a passive sentence⁵.

- (35) a. Umugabo y-iicar-i-w-é-ho íntebe n'=uúmwáana.
man 1SM-sit-APPL-BEN-PFV-LOC chair by=child
'For the man was sat on the chair by the child.'
- b. Íntebe y-iicar-i-w-é-ho umugabo n'=uúmwáana.
chair 1SM-sit-APPL-BEN-PFV-LOC man by=child
'The chair was sat on for the man by the child.'
- (McGinnis and Gerdtz 2004:p. 162)

Accounting for this symmetry, they propose that HighApplP forms a phase and has a phase-EPP feature, which is satisfied by movement of some element within the phasal domain to its Spec position. This phase-EPP feature is crucially not satisfied by External Merge as the element must come initially from a position within the domain of a phase. The object symmetries between the argument in SpecHighAppl and the VP complement are derived as in (36), where the DO is attracted to the outer SpecHighApplP to check its phase-EPP feature. Both IO and DO are now in SpecHighApplP and accessible to higher operations.

⁵Note that *-i-* is glossed as BEN in this example following that of McGinnis and Gerdtz (2004), but it is at least phonologically indistinct from the APPL in other examples with applied EPCs. I contend these are in fact the same morpheme.



In the remainder of the paper, I show how this accounts for the differences in object symmetry between applied and nuclear EPCs. First, I show how nuclear EPCs display object symmetry in a LowApplP. Following that, I argue that the object asymmetries in applied EPCs are derived naturally following the same principles and claims from McGinnis and Gerdtz (2004) and McGinnis (2001, 2000). Although their proposal explains symmetrical object properties with high applicatives, I show that possessor raising to SpecHighApplP, in fact, derives the opposite.

4.2 Nuclear EPCs

In McGinnis's (2000) proposal, LowApplP (ApplIP in her terms) does not constitute a phase. This accounts for object asymmetries in a variety of languages' double object constructions. Given there is no phase-EPP feature associated with LowApplP, only the higher argument below VP can move to the escape hatch position at SpecVP to check some EPP features and leave the phase.

In Kinyarwanda, however, double object constructions have apparently symmetrical objects. This is seen in the set of sentences below. The IO in (37b) and DO in (37c) can both be the subject of a passive. Other notable tests of symmetry reveal that IOs and DOs can also be both incorporated, relativized, and have flexible word ordering (Jerro 2015, Gary and Keenan 1977, Kimenyi 1980; among others).

- (37) a. Umukoôbwa y-a-bwii-ye umuhuûngu ibinyóma.
 girl 1SM-PST-tell-PFV boy lies
 'The girl told lies to the boy.'
- b. Umuhuûngu y-a-bwii-w-e ibinyôma n'=úmukoôbwa.
 boy 1SM-PST-tell-PASS-PFV lies by=girl
 'The boy was told lies by the girl.'
- c. Ibinyóma by-a-bwii-w-e umuhuûngu n'=úmukoôbwa.
 lies 8SM-PST-tell-PASS-PFV boy by=girl

‘Lies were told to the boy by the girl.’
(Kimenyi 1980:p. 133)

Preserving McGinnis’ analysis of LowApplP not constituting a phase, I propose that symmetry between arguments below VP can be accounted for by the simple stipulation that LowAppl undergoes head-movement to V. In doing this, both arguments in the specifier and complement position of LowApplP become equidistant from V, following Chomsky’s (1995) definition of equidistance in (38).

- (38) If α , β are in the same minimal domain, they are equidistant from γ .
(Chomsky 1995:p. 169)

Both possessor and possessum are then potential goals to raise to SpecVP to escape its phase. This assumption that LowAppl undergoes head-movement to V accounts for not just the object symmetry between possessor and possessum, but the object symmetry between IO and DO more generally. In any case, given the analysis advocated for above, it is clear that however object symmetry between recipient/goal and theme, for instance, arises, it must be the same way that such symmetries arise for possessor-possessum.

4.3 Applied EPCs

Recall that McGinnis’ phase-EPP feature is able to account for symmetry between DOs and AOs by triggering movement of the DO to HighApplP phase-edge, making it accessible to operations outside of the phase. This proposed phase-EPP feature moves an element that is within the phasal domain to its outer specifier, while the inner specifier is occupied by an Externally Merged item. The inner specifier is filled by a base-generated element, preventing it from satisfying the phase-EPP feature, as it is not from within the phasal domain.

This explains why the DO is able to escape the HighApplP phase to show object properties alongside the AO. In my analysis of applied EPCs, there is nothing externally merged in SpecHighApplP to begin with. There is simply a phase-EPP feature that attracts the highest argument. After this phase-EPP feature is valued, the phase is complete and inaccessible to outside operations.

This requires an initial assumption that HighApplP, which is generally considered to have the function of introducing an argument, can project without having anything Externally Merge in its specifier position. This is not a baseless assumption in Kinyarwanda. Jerro (2023) and Beavers, Jerro, and Koontz-Garboden (2015) note that the applicative suffix in Kinyarwanda may appear on lexically ditransitive verbs with no apparent change in valency, as in (39).

- (39) a. Karemera y-a-tey-e ibuye Nkusi.
Karemera 1SM-PST-throw-PFV rock Nkusi
‘Karemera threw the rock at Nkusi.’
b. Karemera y-a-ter-ey-e ibuye Nkusi.
Karemera 1SM-PST-throw-APPL-PFV rock Nkusi

‘Karemera threw the rock to Nkusi.’
(Jerro 2023)

The difference between (39a) and (39b) is supposedly an extra thematic role associated with *Nkusi* in (39b). Only in (39b) is *Nkusi* interpreted as a recipient, where a transfer of possession is entailed. Crucially, the number of arguments does not change between sentences. This is to say that there is independent evidence that HighApplP is able to project with an empty specifier, leaving space for an external possessor in our analysis.

With this in mind, we can propose the structure in (40b) to derive EPCs such as (40a).

- (40) a. Ingurube z-a-ri-ir-iye abaána ibíryo.
pigs 10SM-PST-eat-APPL-ASP children food
‘The pigs ate the children’s food’
- b.
-

I stipulate that the possessor begins in SpecDP, with the possessed NP as the complement of D. In this configuration, there are essentially two options; the entire DP moves, or only the embedded possessor DP moves. In the latter case, the possessor moves to SpecHighApplP, potentially receiving an additional θ -role of affectee. Upon reaching the left edge of the phase, the phase-EPP feature is checked, and the phase is closed off to outside operations. This effectively traps the lower NP in the phase domain, making it unable to raise to subject of a passive or be incorporated.

With this phase-based approach, we are also able to explain restrictions on $\bar{A}$ -movement that earlier accounts struggle with. If these asymmetries were due to the Minimal Link Condition, we couldn’t explain why the possessum can’t undergo $\bar{A}$ movements like relativization. As noted in section 2, $\bar{A}$ -movement prevents an issue for both RG and binding accounts of applied EPCs in Kinyarwanda. By trapping the possessum inside a phase, any and all movement is barred,

not just movements associated with any particular privileged grammatical relation or structural position.

5 Conclusion

External possession in Kinyarwanda is complex and shows different properties from other form of valency increasing operations. The manifestation of EPCs in two distinct ways, with vastly different behaviors and semantic conditions, raises questions about the appropriateness of considering applied and nuclear EPCs as variants of the same phenomenon. Further investigation into the semantics of both constructions is likely needed to determine how closely related they are in actuality.. For instance, I have operated under the assumption that the possessor and possessum in nuclear EPCs must have a part-whole relationship, and that the applied possessor must be affected by the event following Kimenyi (1980) and Van de Velde (2020). However, in my knowledge there have been no studies directly considering the semantics of these constructions, and much of the corpus data lacks the necessary context to determine if these conditions universally hold. Assessing the validity of affectedness conditions imposed on applied external possessors is crucial for future research and holds implications for the advocated analysis, particularly concerning the assumptions made regarding thematic raising.

Discourse properties of applied and nuclear EPCs are also of essential importance. Many Bantu languages have word order and other syntactic properties that are highly influenced by pragmatic factors and information-structure concerns, with some arguing that Bantu languages have a much more complex system of agreement with discourse-related features or lack Case entirely and have arguments licensed solely based on discourse properties (Carstens and Diercks 2009, Buell, Riedel, and van der Wal 2011, Morimoto 2000, Carstens and Mletshe 2015; among others). In this case, the analysis proposed here would likely require broad revisions. It is possible, and potentially even likely, that EPCs in Kinyarwanda are the result of information-structure constraints determined by topicality, focus, and backgroundedness. This may account for properties such as the inability of external possessors to appear as emphatic pronouns (Kimenyi 1980:p. 102-103) as in (41)⁶, or requirements that the external possessor be above the

⁶ Note that although Kimenyi claims this to be the case in both applied and nuclear EPCs, it is interestingly possible to have (potentially) in-situ wh-phrases representing external possessors in nuclear (1) but not applied EPCs (2). In this case, however, the wh-phrase external possessor must follow the possessum rather than precede it, as in (1b).

- (1) a. * Umugóre y-a-shoko-je nde umusatsi?
 woman 1SM-PST-comb-PFV who hair
 b. Umugóre y-a-shoko-je umusatsi nde?
 woman 1SM-PST-comb-PFV hair who
 ‘Whose hair did the woman comb?’
- (2) a. * Umukoóbwa y-a-som-i-ye nde igitabo?
 girl 1SM-PST-read-APPL-PFV who book
 b. * Umukoóbwa y-a-som-i-ye igitabo nde?
 girl 1SM-PST-read-APPL-PFV book who

possessum on the referentiality hierarchy (Van de Velde 2020). Nevertheless, these issues remain a point for further research as it clear the significance of information-structure concerns in any study of Bantu.

- (41) a. * Umuhûngu y-a-som-e-ye twê ibitabo.
 boy 1SM-PST-readAPPL-PFV us books
 Intended: ‘The boy read our books.’
 b. Umuhûngu y-a-du-som-e-ye ibitabo.
 boy 1SM-PST-2PL.OM-readAPPL-PFV books
 ‘The boy read our books.’
 (Kimenyi 1980:p. 102-103)

The analysis advocated here attempts to explain several differences in how object symmetry is observed in Kinyarwanda EPCs by proposing that they are derived via different but related means. Nuclear EPCs are seen showing the same level of object symmetry as in standard ditransitive verbs/ DOCs. As such, I have proposed they are derived through a LowApplPAT in the sense advocated by Cuervo (2003), denoting a static relationship between two individuals. With the possessor in the specifier and possessum as complement of LowAppl, we are able to derive object symmetries on the assumption that there is head movement of LowAppl to V, thus making both arguments equidistant to SpecVP and enabling them to escape the phase.

In contrast, applied EPCs seemingly involve a stripping of several object properties away from the possessum. Possessums are unable to undergo both A-movement (i.e. passivization, incorporation) and $\bar{A}$ -movement (i.e. relativization, clefting, dislocation). I have proposed applied EPCs are derived via movement to the specifier of a HighApplP, motivated by observations about word order, island effects, and the external possessor’s ability to associate with arguments above VP. Following from the analysis of possessor raising, we naturally derive the inability of the possessum to undergo any movement by following McGinnis’s (2000, 2001) proposal that HighAppl constitutes a phase. Upon checking its phase-EPP feature, the phase is closed off and the possessum remains trapped, inaccessible to further derivations.

The analysis advocated here is intuitive in that it supposes two (apparent) variants of a construction that appear superficially different are in fact manifest via different means. The explanatory power allows differences in object (a)symmetries fall naturally from the different derivations of EPCs, and we are able to relate them to other superficially similar constructions. I hope to have shown that external possession in Kinyarwanda can be explained with more contemporary theory and that there can be more complexity than the surface may initially show.

-
- c. Umukoóbwa y-a-som-i-ye igitabo cya=nde?
 girl 1SM-PST-read-APPL-PFV book CONN₇=who
 ‘Whose book did the girl read?’
 (Simon Pierre Munyaneza, personal communication)

The ability to appear with the inverse word order is predicted by the analysis of nuclear EPCs given above. I do not have an in-depth explanation for why the possessor in (1b) must follow the possessum, although Ndayiragije’s (1999) proposal for a TP-internal FocP to which wh-words move to Spec of may offer some explanation if we stipulate the possessum moves to some higher position to escape the domain of focus.

References

- Beavers, John, Kyle Jerro, and Andrew Koontz-Garboden. 2015. Roots, applicatives, and templatic meaning in Kinyarwanda ditransitives. In *Presented at Sinn und Bedeutung 23*, 130–145.
- Bickford, J Albert. 1986. Possessor ascension in Kinyarwanda. In *CLS 22*, vol. 22, 129–143.
- Buell, Leston C., Kristina Riedel, and Jenneke van der Wal. 2011. What the Bantu languages can tell us about word order and movement. *Lingua* 121: Movement and Word Order in Bantu, 689–701. <https://www.sciencedirect.com/science/article/pii/S0024384110002561>.
- Carstens, Vicki and Michael Diercks. 2009. Parameterizing case and activity: hyper-raising in Bantu. *Linguistics publications (MU)*.
- Carstens, Vicki and Loyiso Mletshe. 2015. Radical defectivity: implications of Xhosa expletive constructions. *Linguistic Inquiry* 46:187–242.
- Chomsky, N. 1995. *The minimalist program* Current studies in linguistics series. MIT Press. <https://books.google.ca/books?id=vtPQiYCNpjgC>.
- Chomsky, Noam and Michael Kenstowicz. 1999. Derivation by phase. *An Annotated Syntax Reader* 482.
- Cuervo, Maria Cristina. 2003. Datives at large. Doctoral dissertation, <http://dspace.mit.edu/handle/1721.1/7991>.
- Davies, William D. 1997. *Relational succession in Kinyarwanda possessor ascension*.
- Deal, Amy Rose. 2017. External Possession and Possessor Raising. In *The Wiley Blackwell Companion to Syntax, Second Edition*, 1–32. John Wiley & Sons, Ltd. <https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118358733.wbsyncom047>.
- Dryer, Matthew S. 1983. Indirect objects in kinyarwanda revisited. *Studies in relational grammar* 1:129–140.
- Gary, Judith Olmsted and Edward Louis Keenan. 1977. On collapsing grammatical relations in universal grammar. In *Grammatical relations*, 83–120. Brill.
- Hornstein, Norbert. 1999. Movement and Control. *Linguistic Inquiry* 30:69–96. <https://doi.org/10.1162/002438999553968>.
- Jerro, Kyle. 2015. Revisiting object symmetry in Bantu. In *Selected proceedings of the 44th annual conference on african linguistics*, 130–145.
- Jerro, Kyle. 2023. The semantics of applicativization in Kinyarwanda. *Natural language and linguistic theory*.

- Kimenyi, Alexandre. 1980. A relational grammar of Kinyarwanda. Doctoral dissertation,
- McGinnis, Martha. 2000. Phases and the syntax of applicatives. In *North east linguistics society*, vol. 31, 7.
- McGinnis, Martha. 2001. Variation in the phase structure of applicatives. *Linguistic Variation Yearbook* 1:105–146.
- McGinnis, Martha and Donna B. Gerds. 2004. A phase-theoretic analysis of Kinyarwanda multiple applicatives. In *The 2003 canadian linguistic association annual conference*, 154–165. Université du Québec à Montréal. https://explore.openaire.eu/search/other?orpId=datacite____:fc2a07abe281b9750d06174e3a7f2852.
- Morimoto, Yukiko. 2000. *Discourse configurationality in Bantu morphosyntax*. Stanford University.
- Ndayiragije, Juvénal. 1999. Checking economy. *Linguistic Inquiry* 30:399–444. <http://www.jstor.org/stable/4179070> (27 Jan., 2023).
- Pylkkänen, Liina. 2000. What applicative heads apply to. *University of Pennsylvania Working Papers in Linguistics* 7:18.
- Simango, Silvester Ron. 2007. Enlarged arguments in Bantu: evidence from Chichewa. *Lingua* 117:928–949. <https://dx.doi.org/10.1016/j.lingua.2006.05.003>.
- Tyler, Matthew. 2021. Two kinds of external possession in Mississippi Choctaw. *Syntax* 24:78–122. <https://onlinelibrary.wiley.com/doi/abs/10.1111/synt.12205>.
- Van de Velde, Mark. 2020. Concernee-concern constructions: a comparative study of external possession in the Bantu languages. *Studies in language* 44:70–94.

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